

# Solid Mechanics CA1 & CA2, TME235

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## 1. Computer Assignment 1:

### 1.1 Problem Statement:

A clamped steel beam with a rectangular cross-section is used to carry a person standing at its free end. The beam is modelled using Euler-Bernoulli and Timoshenko Beam theories for analytical solution and FEM is conducted for the same using Abaqus and CalFEM in python programming language. For two beam lengths i.e.  $L_1 = 2\text{m}$  and  $L_2 = 0.25\text{m}$ , the following tasks are performed:

- Determine the deflection  $w(x)$  along the length of the beam considering both applied point load and the self-weight of the beam.
- To calculate the normal stress  $\sigma_{xx}$ ,  $(x, -h/2)$  and von-mises stress  $\sigma_{eM}$ ,  $(x, -h/2)$  to assess the likelihood of yielding.
- To compare and discuss the results from different methods along with a mesh convergence study for FEM models.
- Discussion on design improvement to avoid stress concentrations in the beam.

Material Properties:  $E = 200 \text{ GPa}$ ,  $\nu = 0.3$ ,  $\rho = 7800 \text{ kg/m}^3$ ,  $\sigma_y = 450 \text{ Mpa}$ ,  $g = 9.81 \text{ m/s}^2$ .

### 1.2 Euler Bernoulli Beam Model:

#### 1.2.1 Governing Equation and Boundary Conditions:

Euler-Bernoulli beam theory is one of the most widely used theories for calculation of deflection, stresses and strains when a beam is under a transverse loading. In our assignment we assume the material to be linear elastic. So, for linear elastic solids, the theory states that:

$$EIw'''' = -q \quad (1.1)$$

$$M(x) = -EIw''(x) \quad (1.2)$$

$$V(x) = M'(x) \quad (1.3)$$

Using equation (1.1), (1.2) and (1.3) we will solve for deflection  $w(x)$ , Normal stress  $\sigma_{xx}$ , and Von-Mises Stress  $\sigma_{eM}$ .

For Deflection, integrate eqn (1.1) four time with respect to x, we get

$$w(x) = 1/EI \left( \int \int \int \int (-q) dx dx dx dx + \frac{C_1 x^3}{6} + \frac{C_2 x^2}{2} + C_3 x + C_4 \right) \quad (1.4)$$

Where  $C_1, C_2, C_3, C_4$  are integral constants which can be determined using boundary conditions,  $q$  is load per unit length (self-weight in our case),  $E$  is Young's Modulus and  $I$  is Moment of Inertia.

Boundary Conditions for the beam is as follows:

At  $x = 0$ , where beam is clamped

$$w(0) = 0$$

$$w'(0) = 0$$

Where  $w'(x)$  represents slope/rotation of the beam.

At  $x = L$  where person is standing on the beam

$$M(L) = 0$$

$$V(L) = -P$$

Where  $P$  is the point load (Person's weight) at the end of the beam.

For Normal stress  $\sigma_{xx}$  we used equation (2.13) from lecture notes:

$$\sigma_{xx} = -Ez w''(x)$$

For Von-mises stress  $\sigma_{eM}$  we used equation (1.35) from lecture notes:

$$\sigma_{eM} = \sqrt{\frac{3}{2} \sigma_{ij}^{dev} \sigma_{ij}^{dev}}$$

Where  $\sigma_{ij}^{dev}$  is the Deviatoric Stress.

## 1.2.2 Methodology and Results:

To solve equations and boundary condition mentioned in section 1.2.1 we used python as seen in the code snippet below.

*Python Code Snippet 1:*

#Variables

p,q,x,L,Emod,I,C1,C2,C3,C4,z = symbols ('p q x L Emod I C1 C2 C3 C4 z',  
real=True)

```

# Equation

w1 = (integrate(-q,x,x,x,x)+C1*x**3/6+C2*x**2/2+C3*x+C4)/(Emod*I)

w1prime = diff(w1,x)

M1 = -simplify(Emod*I*diff(w1prime,x))

V1 = simplify (diff(M1,x))


#Boundary Conditions

eqns1 = [ w1.subs(x,0), w1prime.subs(x,0),M1.subs(x,L), V1.subs(x,L)+p]


#Solution

unknowns1 = (C1,C2,C3,C4)

sol1 = solve (eqns1,unknowns1)

w_total = simplify (w1.subs(sol1))

```

Using Python, we found deflection at  $x = L$  is around 33.81 mm and 0.0597 mm for the beam with length 2m and 0.25 m respectively.

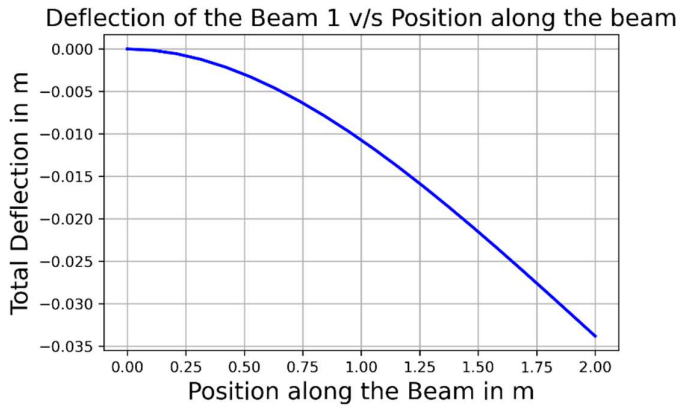


Fig 1(a) Deflection of beam 1 i.e. 2m in length

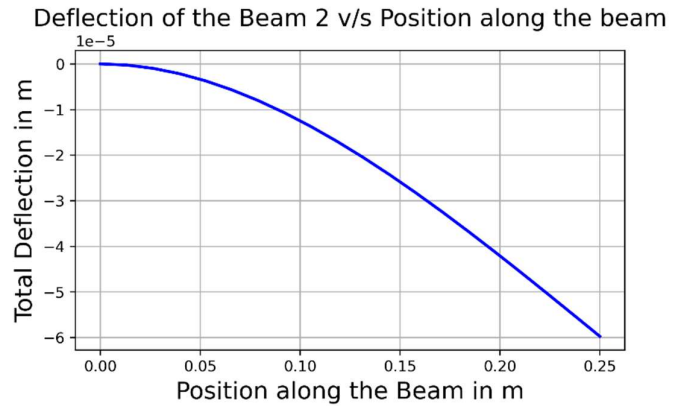


Fig 1(b) Deflection of Beam2 i.e. 0.25m in length

Fig 1 Deflection of the Beams in Euler-Bernoulli Beam Models

It was also observed that the stress concentration in the beam was developed at the clamping region as beam experience's reaction forces and moments due to applied load.

Normal stress  $\sigma_{xx}$  is computed at  $z = -h/2$  i.e. bottom of the beam, along the length of the beam and it is shown in Figure 2.

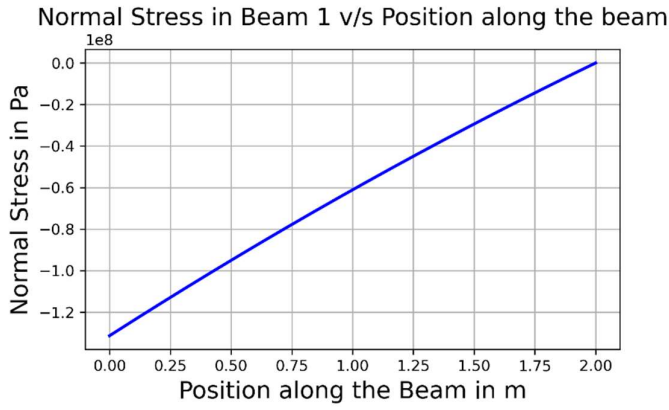


Fig 2(a) Normal Stress in beam 1 i.e. 2m in length

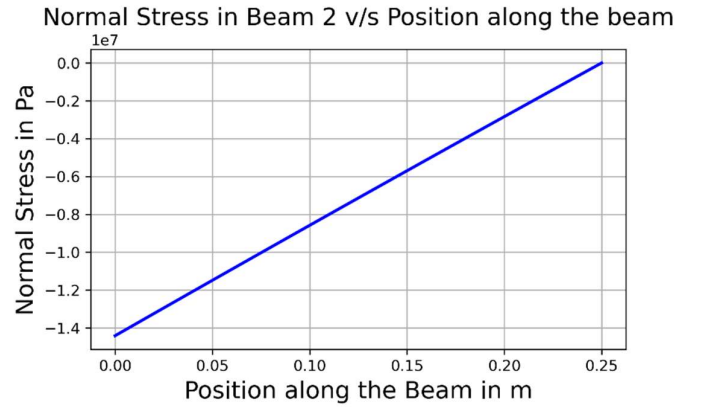


Fig 2(b) Normal Stress in Beam2 i.e. 0.25m in length

Fig 2 Normal Stress in Euler-Bernoulli Beam Models

Similarly, Von-Mises Stress  $\sigma_{eM}$  was also computed at  $z = -h/2$  to check yielding criterion. A beam is considered to be yielding if,

$$\sigma_{eM} > \sigma_y$$

But as computed, in this case beams are not yielding as  $\sigma_{eM} < \sigma_y$ . A graph has been plotted between von-mises stress and length of the beam in Figure 3.

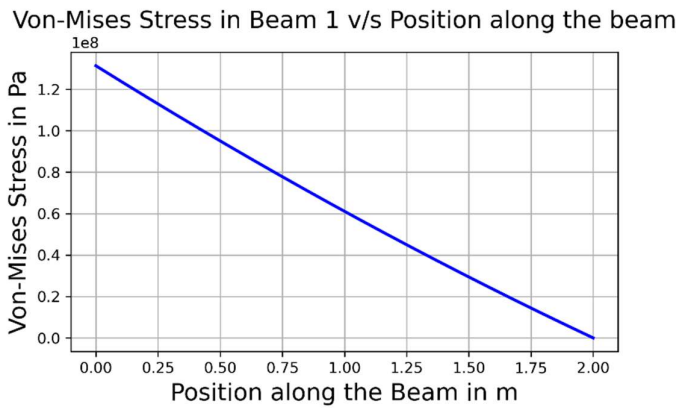


Fig 3(a) Von-Mises Stress in Beam 1 i.e. 2m in length

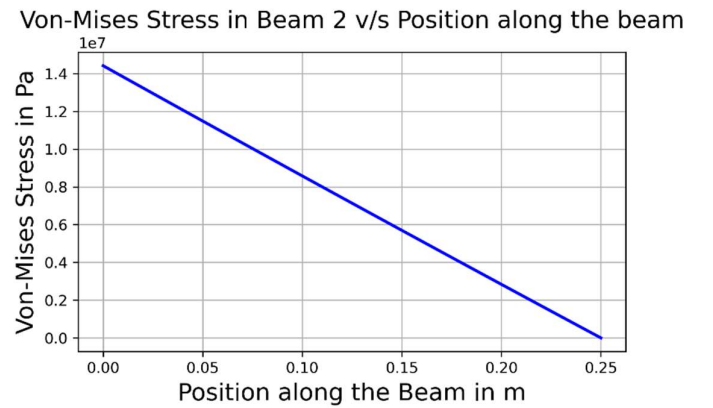


Fig 3(b) Von-Mises Stress in Beam 2 i.e. 0.25m in length

Fig 3 Von-Mises Stress in Euler-Bernoulli Beam Models

## 1.3 Timoshenko Beam Model:

### 1.3.1 Governing Equation and Boundary Conditions:

In section 1.2 we saw the implementation of Euler-Bernoulli theory for calculation of deflection, normal stress and von-mises stresses. But in this section, we will use Timoshenko Beam model for solving deflection, normal stresses and von-mises stresses, and major difference between the two models are Euler-Bernoulli model doesn't take into account the shear stress in the beam as opposed to Timoshenko where shear stress is considered in the calculation. So, we will compute Timoshenko beam model and then we can compare the values with Euler-Bernoulli beam model.

For linear elastic solids, the theory states that

$$EI\phi'''(x) = -q \quad (1.5)$$

$$\phi = \frac{1}{EI} \left( \frac{-qx^3}{6} + \frac{C_1x^2}{2} + C_2x + C_3 \right) \quad (1.6)$$

$$M(x) = -EI\phi'(x) \quad (1.7)$$

$$V(x) = M'(x) \quad (1.8)$$

$$w(x) = \left( -\frac{EI}{GK_sA} \right) \phi' + \int \phi dx + C_4 \quad (1.9)$$

Using equations (1.5), (1.6), (1.7), (1.8) and (1.9) we will solve for deflection  $w(x)$ , normal stress  $\sigma_{xx}$  and von-mises stress  $\sigma_{eM}$ .

Before solving for deflection, we must first solve for integral constants  $C_1, C_2, C_3, C_4$ , and that will be done using the above equations and boundary condition. Following are the boundary conditions for our case in Timoshenko beam theory.

At  $x = 0$ , where beam is clamped

$$w(0) = 0$$

$$\phi(0) = 0$$

At  $x = L$ , free end of the beam

$$M(x) = 0$$

$$V(x) = -P$$

Using these boundary conditions, we will solve for integral constants and we will use equation (1.9) for determining deflection in the beam.

For normal stress  $\sigma_{xx}$  we use equation (3.28) from lecture notes and Hooke's law we get

$$\sigma_{xx} = -Ez\phi'(x)$$

For Von-Mises Stress  $\sigma_{eM}$  we use equation (1.35) from lecture notes

$$\sigma_{eM} = \sqrt{\frac{3}{2} \sigma_{ij}^{dev} \sigma_{ij}^{dev}}$$

### 1.3.2 Methodology and Results:

To solve the equation mentioned in the above section we used python as seen in the code below.

*Python Code Snippet 2:*

#Variables

```
p,q,x,L,Emod,I,C1,C2,C3,C4,z,Gmod,phi,K_s,A = symbols('p q x L Emod I C1 C2 C3 C4 z Gmod phi K_s A', real=True)
```

# Equation

```
phi1 = (-q*x**3/6+C1*x**2/2+C2*x+C3)/(Emod*I)
```

```
M1 = -simplify (Emod*I*diff (phi1, x))
```

```
V1 = simplify (diff (M1, x))
```

```
w1 = (M1/(Gmod*K_s*A)) +integrate (phi1, x) +C4
```

#Boundary Conditions

```
eqns1 = [w1.subs(x,0),phi1.subs(x,0),M1.subs(x,L),V1.subs(x,L)+p]
```

#Solution

```
unknowns1 = (C1,C2,C3,C4)
```

```
sol1 = solve(eqns1,unknowns1)
```

```
w_total TT = simplify(w1.subs(sol1))
```

Using Timoshenko beam model, we found that deflection at  $x = L$  is around 33.83 mm and 0.062 mm for the beam with length 2m and 0.25 m respectively.

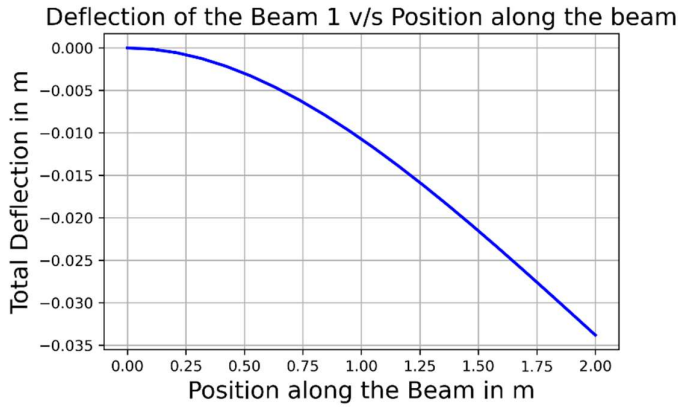


Fig 4(a) Deflection of beam 1 i.e. 2m in length

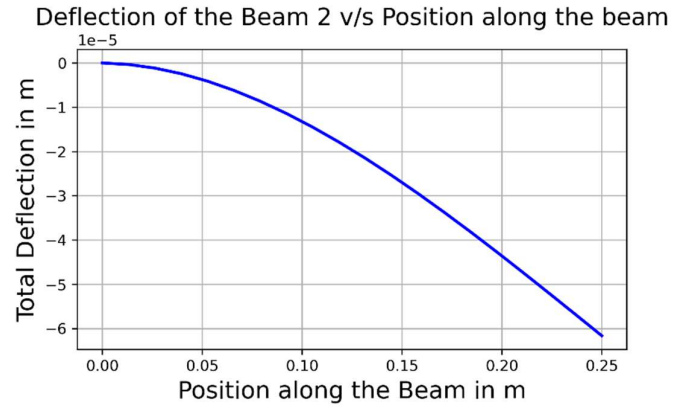


Fig 4(b) Deflection of Beam2 i.e. 0.25m in length

Fig 4 Deflection of the Beams in Timoshenko Beam Models

Similar to Euler-Bernoulli's Beam Model, stress concentration is developed at the clamped area as it develops a resisting force to counter the applied force on the beam. Normal stress is computed at  $z = -h/2$  along the length of the beam as shown in figure 5. Normal stress at  $(x = 0, z = -h/2)$  are 131.37 Mpa and 14.41 Mpa in beam 1 and beam 2 respectively.

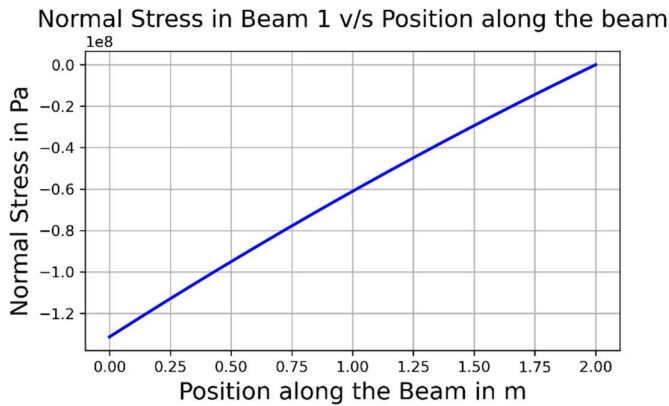


Fig 5(a) Normal Stress in beam 1 i.e. 2m in length

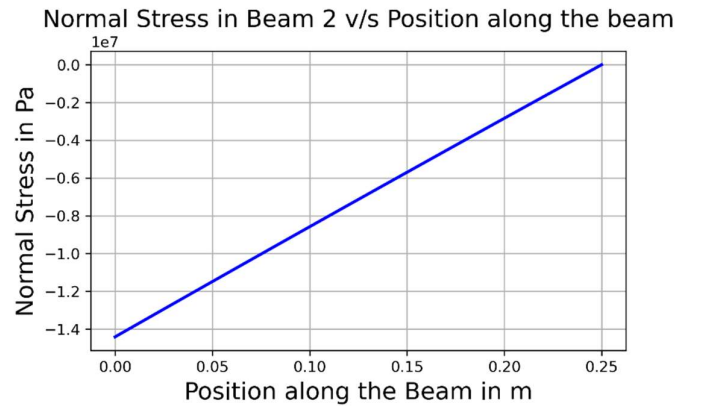
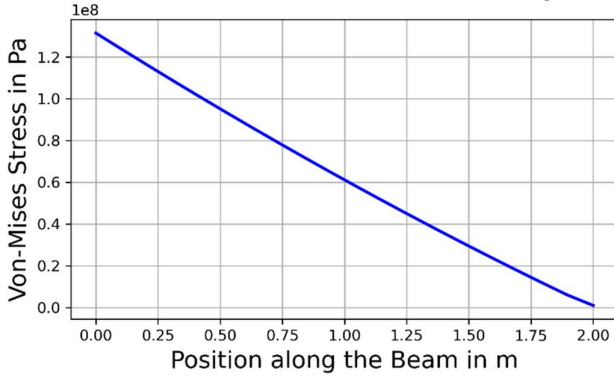


Fig 5(b) Normal Stress in Beam2 i.e. 0.25m in length

Fig 5 Normal stress in Beams in Timoshenko Beam Models

Similarly Von-Mises Stress is also computed at  $z = -h/2$  along the length of beam. Von-Mises stress at  $(x = 0, z = -h/2)$  are 131.38 Mpa and 14.45 Mpa in beam1 and beam2 respectively.

Von-Mises Stress in Beam 1 v/s Position along the beam



Von-Mises Stress in Beam 2 v/s Position along the beam

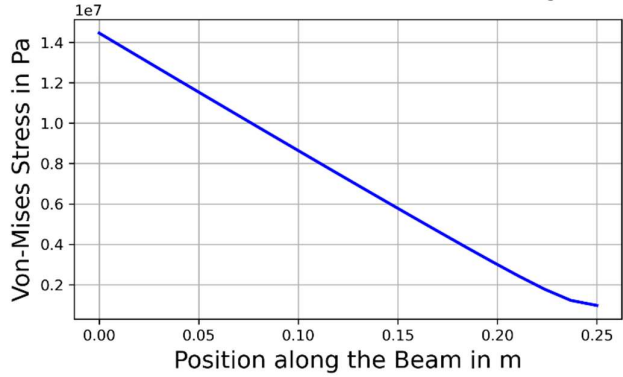


Fig 6(a) Von-Mises Stress in beam 1 i.e. 2m in length

Fig 6(b) Von-Mises Stress in Beam2 i.e. 0.25m in length

Fig 6 Von-Mises's stress in Beams in Timoshenko Beam Models

### 1.4 Mesh Convergence in FEM Models:

In the 2D analysis of a cantilever beam, we performed a mesh convergence study for both Abaqus and CALFEM to validate the results. The primary goal of this study was to ensure that the accuracy of the simulation results was not compromised by the mesh size and that further refinement would no longer significantly impact the output. Both software packages used quad elements, which are well-suited for this type of structural problem due to their balance of accuracy and efficiency.

Through this process, we observed that the results began to converge at  $0.002\text{ m}$  in Abaqus and  $0.0031\text{ m}$  in Calfem for Beam 1 and  $0.001\text{ m}$  in Abaqus and  $0.00156\text{ m}$  in Calfem for Beam2. After this point, further mesh refinement in both Abaqus and CALFEM produced negligible differences, confirming the reliability of the solution. This convergence ensures that the simulation is both accurate and computationally efficient across different platforms.

Below are the convergence study results. For convergence study, we used deflection in the beam to plot the graph



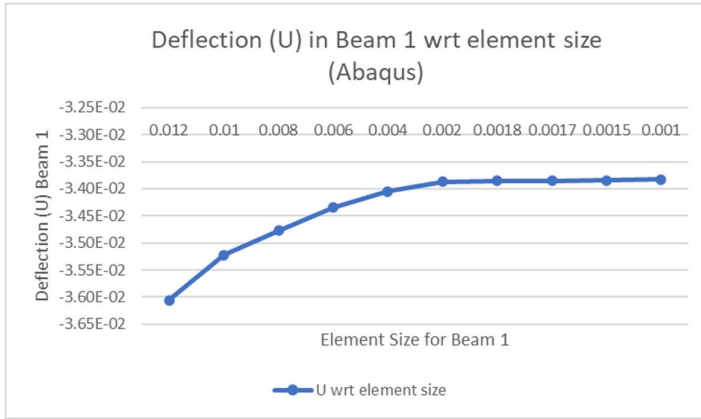


Fig 7(a) Abaqus Mesh Convergence for Beam 1

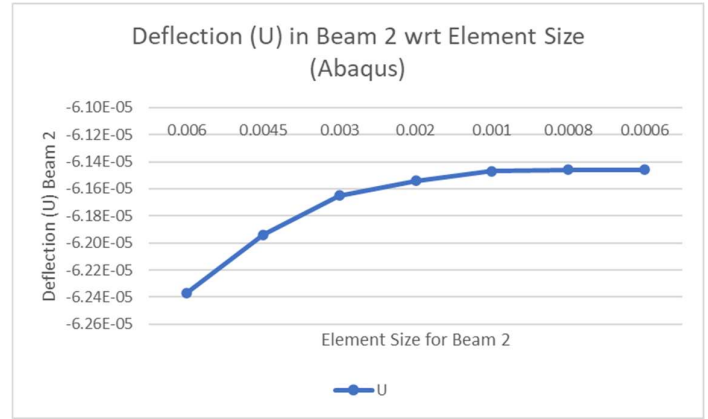


Fig 7(b) Abaqus Mesh Convergence for Beam 2

Fig 7 Abaqus Mesh Convergence for Beams

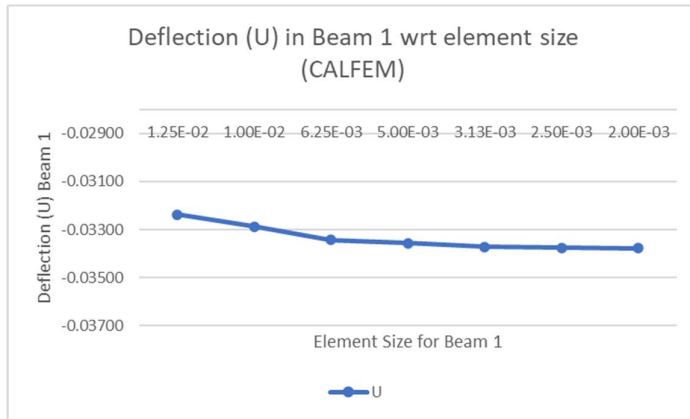


Fig 8(a) Calfem Mesh Convergence for Beam 1

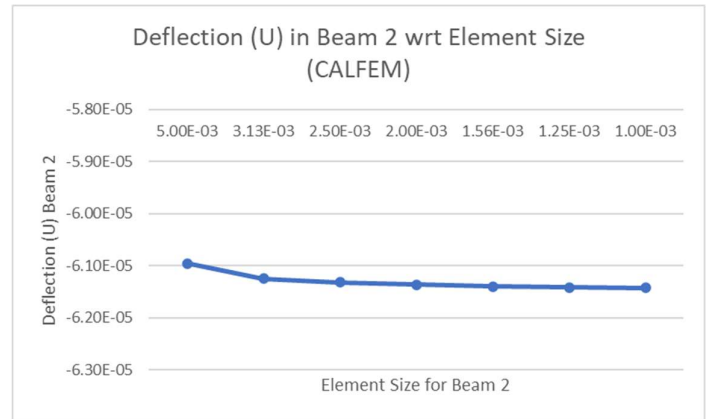


Fig 8(b) Calfem Mesh Convergence for Beam 2

Fig 8 Calfem Mesh Convergence for Beams

## 1.5 Comparison Between Beam Models:

In this section we will compare the results we achieved in Euler-Bernoulli beam model and Timoshenko beam model along with FEM results. Based on the comparison we can discuss which model suits best for solving Beam problems.

### 1.5.1 Deflection:

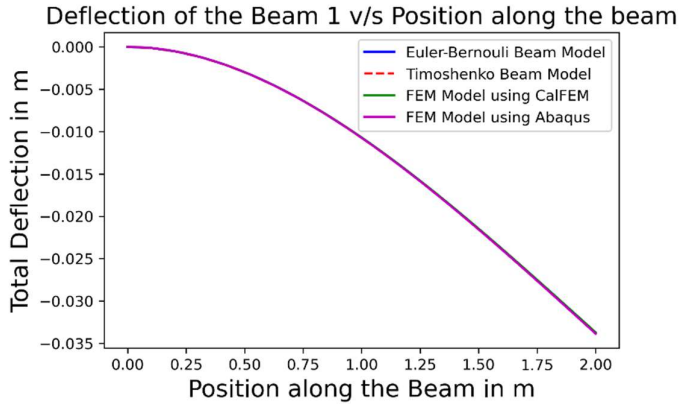


Fig 9(a) Deflection in beam 1 i.e. 2m in length

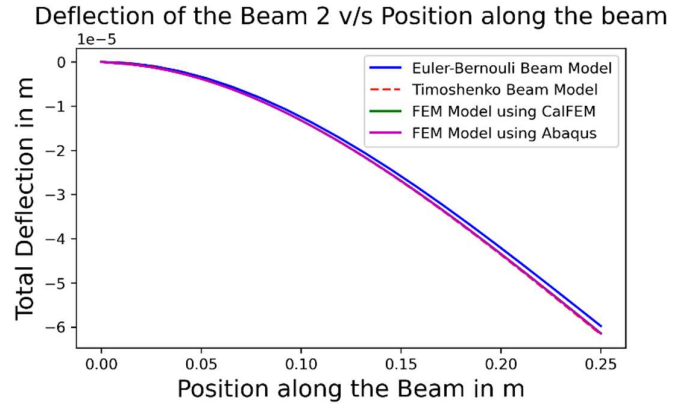


Fig 9(b) Deflection in Beam2 i.e. 0.25m in length

Fig 9 Deflection in Beams in Euler-Bernoulli v/s Timoshenko Beam Models v/s FEM

For Beam 1 when we compare the Euler-Bernoulli and Timoshenko beam model we can see that there is a negligible deviation in the results. This is because in long beam where  $L \gg h, b$  the shear stress developed has negligible effect on the result. So, in a long beam i.e. Beam 1, Timoshenko Beam Model gives similar result to Euler-Bernoulli Beam Model. FEM results also confirm the analytical results we obtained for deflection in beam 1.

But in Beam 2 we can see the deviation in the results between Euler-Bernoulli and Timoshenko Beam model. Here FEM model agrees with the Timoshenko Beam model as shear stress has significant effect on the results in thick beams. So, for shorter beams we can say that Timoshenko Beam Model gives more accurate results compared to Euler-Bernoulli Beam Model.

## 1.5.2 Normal Stress:

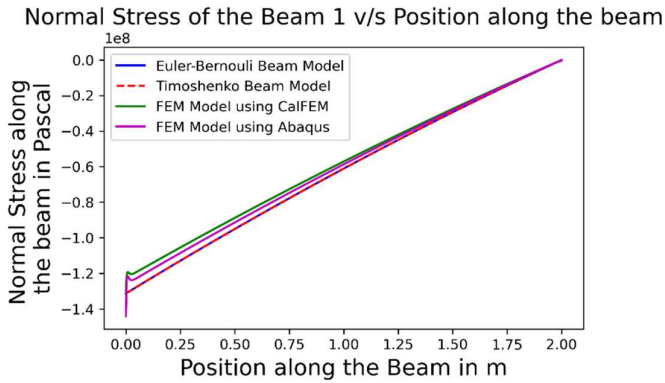


Fig 10(a) Normal Stress in beam 1 i.e. 2m in length

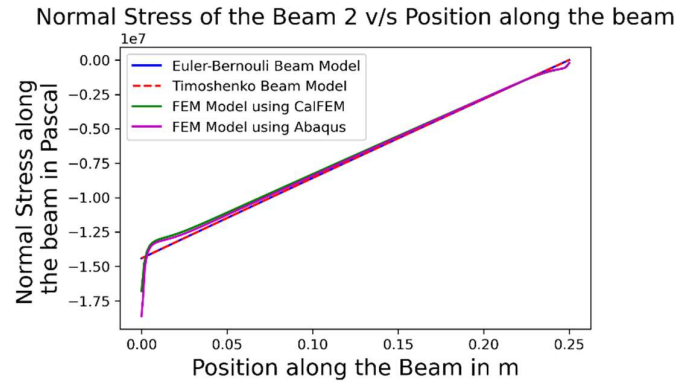


Fig 10(b) Normal Stress in Beam2 i.e. 0.25m in length

Fig 10 Normal Stress in Beams in Euler-Bernoulli v/s Timoshenko Beam Models v/s FEM

For Normal stresses, we can see that there are some deviations between FEM and Analytical solutions. Our assumption is there are some unknown factors which we neglect in our analytical solutions because of which we can see those deviations.

## 1.6 Horizontal Displacement:

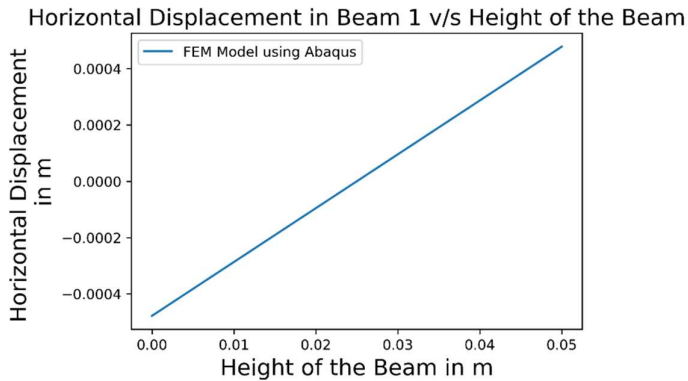


Fig 11(a) Horizontal Displacement in beam 1 i.e. 2m in length

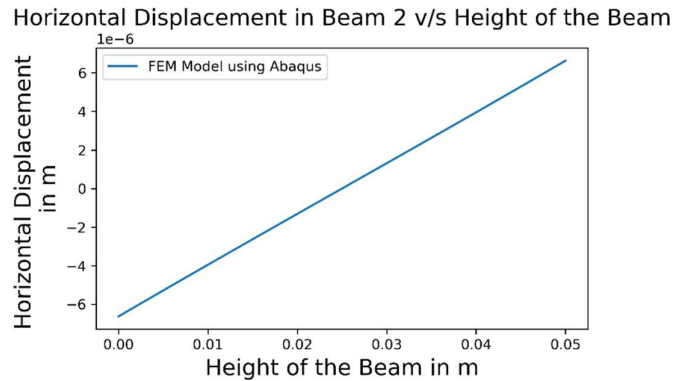


Fig 11(b) Horizontal Displacement in Beam2 i.e. 0.25m in length

Fig 11 Horizontal displacement in Beams

A key assumption for beam kinematics in Euler-Bernoulli beam model was that horizontal displacement varies linearly with respect to  $z$ . We can see that in equation (2.10) in Lecture Notes.

$$u(x, z) = -z \frac{dw}{dx}$$

As per the equation above the displacement should be 0 at  $z = 0$  and should vary linearly away from neutral axis. The graph plotted above confirms the assumption that horizontal displacement varies linearly with respect to  $z$ . So, we can say that kinematic assumption of the beam is very reasonable.

### 1.7 Design Improvements:

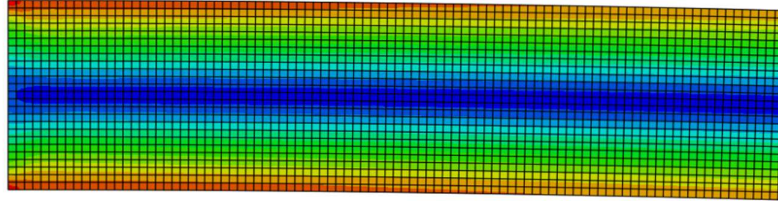


Fig 12 (a) Von-Mises stress at Beam 1

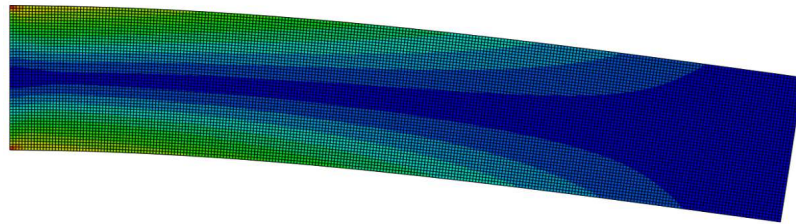


Fig 12 (b) Von-Mises Stress at Beam 2

Fig 12 Contour plot of Von-Mises's stress in the Beams

We can see that there is significant rise in Von-Mises stress at the clamping region which shows stress concentration at these points. There is multiple reason for it one of which can be because of sharp edges at the clamped region. So, we believe the stress concentration at this point can be reduced if we introduce fillet between clamping structures. Fillets can distribute stress more evenly which will help in avoiding such stress concentration areas.

## 2. Computer Assignment 2:

### 2.1 Problem Statement:

We have to compute radial stress  $\sigma_{rr}$  and radial displacement  $u_r$  for an axisymmetric disc where:

Inner radius ( $a$ ) = 100 mm, Outer radius ( $b$ ) = 200 mm, Thickness ( $h_0$ ) = 20 mm,  $E = 200$  GPa,  $\nu = 0.3$  and  $\sigma_{rr}(r = b) = -100$  Mpa

We will be solving this analytically as well as numerically using Python, and then we will update the same code to extract solution for varying thickness of the plate

$$h = \frac{h_0(r - a)}{b - a} + h_0$$

### 2.2 Solution for Axisymmetric with constant thickness:

For Axisymmetric disc following are the equations which will be used to first solve for integral constants  $C_1$  and  $C_2$ .

$$u_r = -\frac{1 - \nu^2}{E} \frac{1}{r} \int r \int f_r dr dr + \frac{C_1 r}{2} + \frac{C_2}{r}$$

$$\epsilon_{rr} = \frac{du_r}{dr}$$

$$\epsilon_{\phi\phi} = \frac{u_r}{r}$$

$$\sigma_{rr} = \frac{E}{1 - \nu^2} (\epsilon_{rr} + \nu \epsilon_{\phi\phi})$$

Boundary conditions for the disc is given below:

- At  $r = a$ ,  $u_r(a) = 0$
- At  $r = b$ ,  $\sigma_{rr}(b) = -100$  Mpa

Using the above equations and boundary conditions we've computed analytical solution for axisymmetric disc as shown in the snippet below:

*Python Code Snippet 3:*

```
#Variables
fr,r,a0,b0,h0,Emod,nu0,C1,C2, P = symbols ('fr r a0 b0 h0 Emod nu0 C1 C2
P',real=True)

#Equation
ur_ = -((1-nu0**2)/(Emod*r))*integrate(r*integrate(fr,r),r)+C1*r/2+C2/r
Strain_rr = diff(ur_,r)
Strain_phi_phi = ur_/r
Sigma_rr_ = (Emod/(1-nu0**2))*(Strain_rr+nu0*Strain_phi_phi)
Sigma_phi_phi = (Emod/(1-nu0**2))*(nu0*Strain_rr+Strain_phi_phi)
```

```

unknowns = (C1,C2)

#Boundary Conditions
eqns = [ur_.subs(r,a),Sigma_rr_.subs(r,b)+P]

#Solution
sol = solve(eqns,unknowns)

```

Similarly we computed radial displacement and radial stress using FEM in Python by using equations from section 5.1.2. Snippet of the code is given below

*Python Code Snippet 4:*

```

r,r1,r2,h0 = symbols('r r1 r2 h0',real=True)
num_elements = 30
num_nodes = num_elements+1
r_values_FEM = np.linspace(a,b,num_nodes)

K_global = np.zeros((num_nodes,num_nodes))
F_global = np.zeros(num_nodes)

D = (E/(1-nu**2))*np.array([[1,nu],[nu,1]])

Ke = []

for i in range(num_elements):
    r_1 = r_values_FEM[i]
    r_2 = r_values_FEM[i+1]
    Ne1 = (r2-r)/(r2-r1)
    Ne2 = (r-r1)/(r2-r1)
    B = Matrix([[diff(Ne1,r),diff(Ne2,r)],[Ne1/r,Ne2/r]])
    K = B.T*D*B*h0*r

    Ke_Matrix = 2*np.pi*integrate(K,(r,r1,r2))
    Ke = Ke_Matrix.subs({r1:r_1,r2:r_2,h0:h})
    K_global[i:i+2,i:i+2]+=Ke

```

It can be seen from the below graphs that the solution we got for analytical and FEM are quite similar for constant thickness of the disc which verifies the computation. So, the radial displacement and radial stress in the disc are 0.044 mm and 135.6 Mpa respectively.

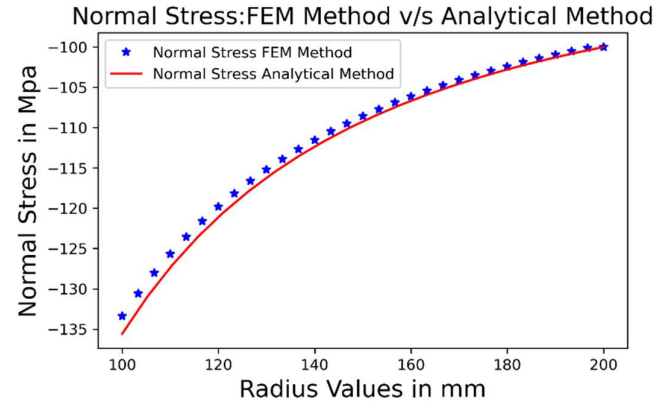
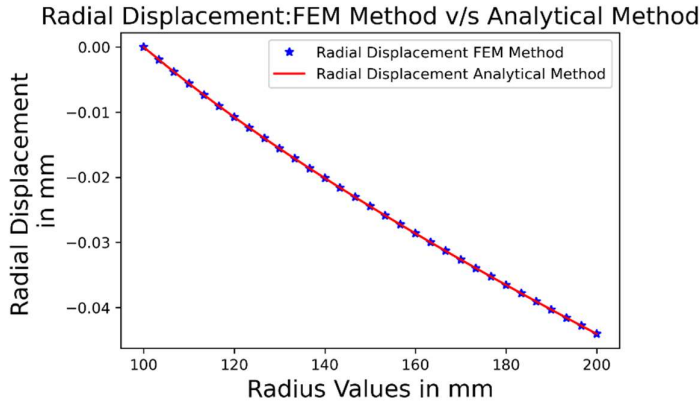


Fig 13(a) Radial Displacement in an Axisymmetric Disc

Fig 13(b) Radial/Normal Stress in Axisymmetric Disc

Fig 13 Solution for Axisymmetric disc with constant thickness  $h_0$

### 2.3 Solution for Axisymmetric disc for variable thickness:

To solve for Axisymmetric disc with variable thickness we used same FEM code that we used in section 2.2 with substituting  $h = h_0 \frac{(r-a)}{b-a} + h_0$ . We could see from the solution that the radial displacement has increased for varying thickness at 0.059 mm. This could be because of increase in internal work as the thickness increased at the free end.

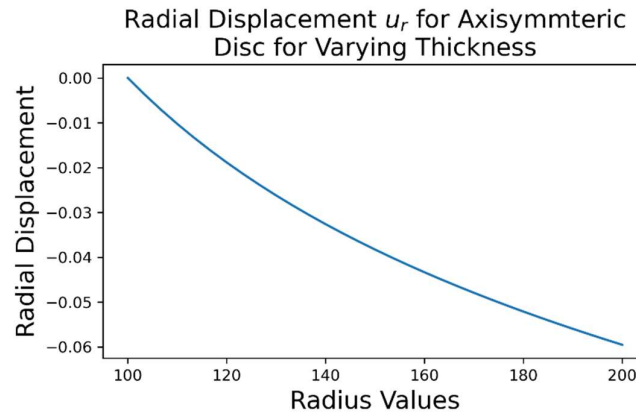


Fig 14 Radial Displacement for Axisymmetric disc for varying thickness